CHAPTER 9

MECHANICAL PROPERTIES OF FLUIDS

CHAPTER ESSENCE AT A GLANCE:
Fluids flow where pressure pushes them! Continuity says fluid speeds up when squeezed into a narrower pipe. Bernoulli discovered that when fluid speeds up, its internal pressure actually DROPS! That pressure drop creates lift under airplane wings!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Hydrostatic & Gauge Pressure:** Absolute pressure at depth h: P = P a + ρ g h. Gauge pressure P g = P - P a = ρ g h. Pressure depends only on depth h, not vessel shape (Hydrostatic Paradox).

• **Pascal's Law & Hydraulic Lift:** Pressure applied to an enclosed fluid transmits undiminished in all directions. Hydraulic Lift: F_1 divided by A_1 = F_2 divided by A_2 implies Output Force F_2 = F_1 (A_2 divided by A_1).

• **Equation of Continuity:** For steady incompressible flow: A_1 v_1 = A_2 v_2 = Constant. Where pipe narrows, fluid speed increases!

• **Bernoulli's Principle:** P + ½ ρ v^2 + ρ g h = Constant. Where fluid speeds up (v), internal pressure drops (P)! Explains airplane wing lift and atomizers.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$P = P_0 + \rho g h, \quad F_B = \rho_{\text{fluid}} V_{\text{disp}} g$$

Hydrostatic Pressure and Archimedes Principle

$$A_1 v_1 = A_2 v_2 \quad \text{(Equation of Continuity)}$$

Conservation of Mass for Incompressible Flow

$$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{Constant}$$

Bernoulli's Equation for Streamline Flow

$$v_t = \tfrac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta}, \quad F_{\text{visc}} = 6\pi\eta r v$$

Stokes' Law and Terminal Velocity

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Many students wrongly assume narrow pipe = high pressure. No! Narrow pipe = high velocity = LOWER pressure (Bernoulli's effect).

• **Alert:** Surface tension tries to minimize surface area, which is why raindrops and soap bubbles naturally pull themselves into perfect spheres!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define surface tension and surface energy. State their SI units. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Surface Tension]** Surface tension (T) is the force per unit length acting perpendicular to an imaginary line drawn on the liquid surface: T = F divided by L (SI unit: N m^{-1}). [1 Mark]

• **[Step 2: Surface Energy]** Surface energy is the work done per unit increase in surface area: U s = W divided by ΔA (SI unit: J m^{-2}). Dimensionally, 1 J m^{-2} = 1 N m^{-1}. [1 Mark]

• **Key Exam Result:** T = F divided by L (N/m), Surface Energy = W divided by ΔA (J divided by m^2)

2. Why do raindrops falling through the atmosphere reach a constant terminal speed instead of accelerating continuously? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Force Balance]** As the raindrop accelerates downward under gravity, the upward viscous drag force F v = 6 π η r v increases with speed. Eventually, the upward forces (viscous drag + buoyant force) exactly equal the downward gravitational weight (mg). Net force becomes zero, so acceleration becomes zero, and the drop falls with a constant terminal velocity v t. [2 Marks]

• **Key Exam Result:** Viscous drag balances gravitational weight (Net F = 0 => constant v t)

3. Derive an expression for the gauge pressure exerted by a liquid column of height h: P = P 0 + ρ g h. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Imaginary Cylinder Forces]** Consider an imaginary liquid cylinder of cross-sectional area A and height h in a liquid of density ρ. Vertical forces acting: 1. Downward force at top face = P 0 A (where P 0 is atmospheric pressure). 2. Upward force at bottom face = P A. 3. Downward weight of cylinder W = m g = (Volume x ρ) g = (A h ρ) g. [1.5 Marks]

• **[Step 2: Equilibrium Condition]** In vertical static equilibrium: P A = P 0 A + A h ρ g. Dividing by area A: P = P 0 + ρ g h. Gauge pressure P - P 0 = ρ g h. [1.5 Marks]

• **Key Exam Result:** Total Pressure P = P 0 + ρ g h, Gauge Pressure = ρ g h

4. Derive an expression for the Terminal Velocity of a small spherical ball of density ρ falling through a viscous fluid of density σ and viscosity η. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Forces on Spherical Body]** Radius r, volume V = 4 divided by 3 π r^3. 1. Downward weight: W = V ρ g = 4 divided by 3 π r^3 ρ g. 2. Upward buoyant force: F B = V σ g = 4 divided by 3 π r^3 σ g. 3. Upward viscous drag: F v = 6 π η r v t. [1.5 Marks]

• **[Step 2: Terminal Velocity Equilibrium]** At terminal velocity v t, net force = 0 => W = F B + F v => 4 divided by 3 π r^3 ρ g = 4 divided by 3 π r^3 σ g + 6 π η r v t. 6 π η r v t = 4 divided by 3 π r^3 (ρ - σ) g. Dividing by 6 π η r: v t = [2 r^2 (ρ - σ) g] divided by (9 η). [1.5 Marks]

• **Key Exam Result:** Terminal Velocity v t = [2 r^2 (ρ - σ) g] divided by (9 η)

5. State and prove the Equation of Continuity for steady streamline flow of an incompressible fluid. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Flow Setup]** Consider a tube of non-uniform cross-section with area A1, velocity v1, density rho1 at inlet, and area A2, velocity v2, density rho2 at outlet. In time Δt: Mass entering inlet Δ m1 = rho1 A1 v1 Δt. Mass leaving outlet Δ m2 = rho2 A2 v2 Δt. [1.5 Marks]

• **[Step 2: Conservation of Mass]** By conservation of mass (fluid cannot accumulate or disappear in steady flow): Δ m1 = Δ m2 => rho1 A1 v1 Δt = rho2 A2 v2 Δt. For an incompressible fluid, density is constant (rho1 = rho2 = ρ): A1 v1 = A2 v2 => A v = Constant. [1.5 Marks]

• **Key Exam Result:** A1 v1 = A2 v2 = Constant

6. State and prove Bernoulli's Principle for the streamline flow of an ideal, incompressible, non-viscous fluid. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** Along a streamline for steady, irrotational, incompressible, non-viscous flow, the sum of pressure energy, kinetic energy, and potential energy per unit volume is constant: $P + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant}$. [1 Mark]

• **[Step 2: Work Done on Fluid Element]** Consider flow from section 1 (area A1, height h1, speed v1, pressure P1) to section 2 (A2, h2, v2, P2). Net work done by pressure forces in time Δt: $W = P1\ A1\ v1\ \Delta t - P2\ A2\ v2\ \Delta t = (P1 - P2)\ \Delta V$ (since $A1\ v1\ \Delta t = A2\ v2\ \Delta t = \Delta V$). [1.5 Marks]

• **[Step 3: Energy Changes & Equating]** Change in kinetic energy: $\Delta K = \text{half } (0.5)\ (\rho\ \Delta V)(v2^2 - v1^2)$. Change in gravitational PE: $\Delta U = (\rho\ \Delta V)\ g\ (h2 - h1)$. By Work-Energy Theorem $W = \Delta K + \Delta U$: $(P1 - P2)\ \Delta V = \frac{1}{2} \rho\ \Delta V\ (v2^2 - v1^2) + \rho\ \Delta V\ g\ (h2 - h1)$. Dividing by ΔV: $P1 - P2 = \frac{1}{2} \rho\ v2^2 - \frac{1}{2} \rho\ v1^2 + \rho\ g\ h2 - \rho\ g\ h1 \Rightarrow P1 + \frac{1}{2} \rho\ v1^2 + \rho\ g\ h1 = P2 + \frac{1}{2} \rho\ v2^2 + \rho\ g\ h2$. [1.5 Marks]

• **Key Exam Result:** $P + \frac{1}{2} \rho\ v^2 + \rho\ g\ h = \text{Constant}$ (Bernoulli's Theorem)

7. (a) Explain the working principle of a Hydraulic Lift using Pascal's law. (2 Marks)n(b) In a hydraulic lift, the diameters of the smaller and larger pistons are 5 cm and 30 cm respectively. What force must be applied to the smaller piston to support a car of mass 1500 kg? (Take g = 9.8 m/s^2) (2 Marks) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Hydraulic Lift Principle]** According to Pascal's law, pressure applied to smaller piston (area A1) is transmitted equally to larger piston (area A2): P = F1 divided by A1 = F2 divided by A2 => F2 = (A2 divided by A1) F1. Since A2 >> A1, a small applied force F1 produces a very large lifting force F2 (Mechanical Advantage MA = A2 divided by A1). [2 Marks]

• **[Step 2: Numerical Calculation]** Smaller radius r1 = 2.5 cm, larger radius r2 = 15 cm. Load on larger piston F2 = m g = 1500 × 9.8 = 14,700 N. F1 = F2 x (A1 divided by A2) = F2 x (r1 divided by r2)^2 = 14700 x (2.5 divided by 15)^2 = 14700 x (1 divided by 6)^2 = 14700 divided by 36 = 408.33 N. [2 Marks]

• **Key Exam Result:** Applied Force F1 = 408.33 N (lifts 14,700 N)